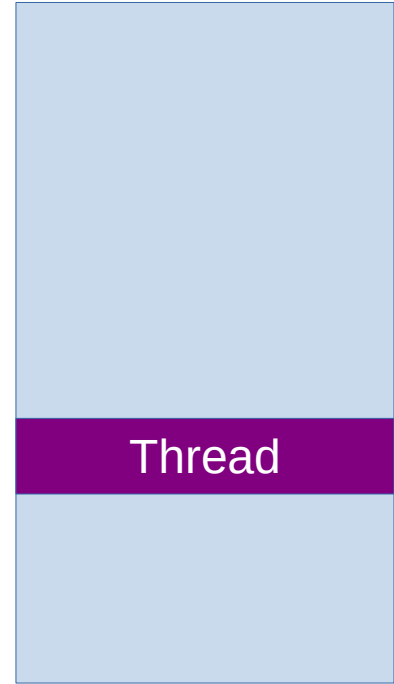
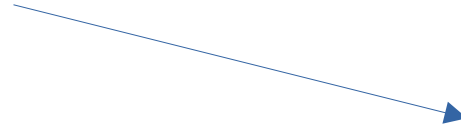


Understanding Threads

What is a thread ?

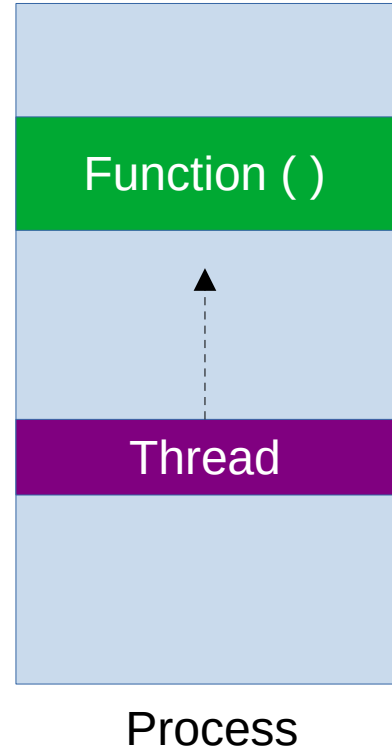
A thread is the smallest unit of execution in a process.



Process

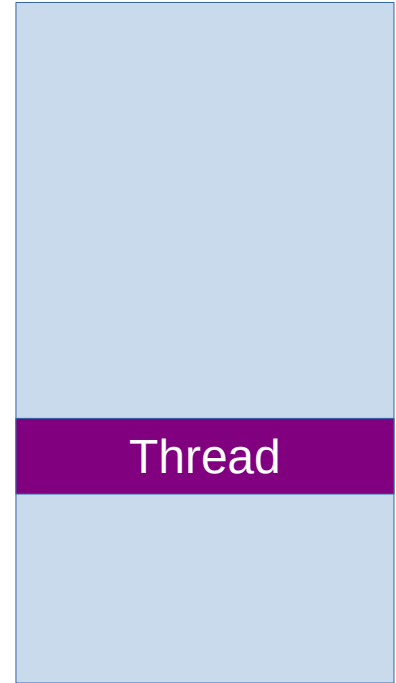
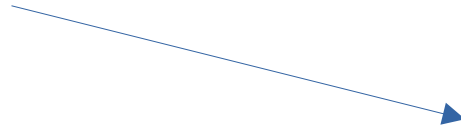
What a Thread does ?

A thread **executes** function code — plain and simple.



How to create Thread ?

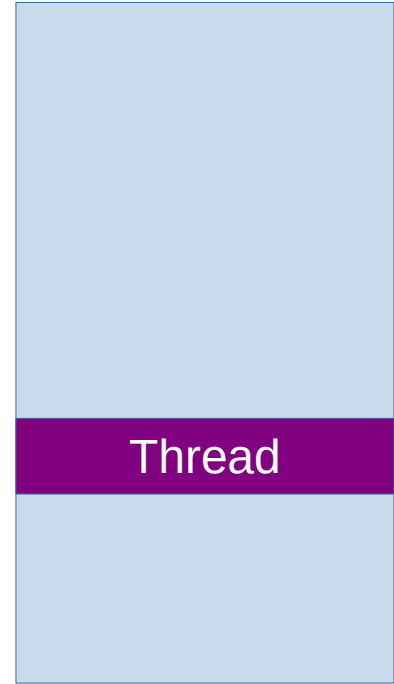
CreateThread()



Process

CreateThread function prototype:

```
HANDLE CreateThread(  
    lpThreadAttributes,  
    dwStackSize,  
    lpStartAddress,  
    lpParameter,  
    dwCreationFlags,  
    lpThreadId  
);
```



Process

CreateThread prototype

```
HANDLE CreateThread(  
    lpThreadAttributes,  
    dwStackSize,  
    lpStartAddress,  
    lpParameter,  
    dwCreationFlags,  
    lpThreadId  
);
```

hThread

Thread

Process



CreateThread prototype

```
HANDLE CreateThread(
```

```
( NULL ) lpThreadAttributes,
```

```
dwStackSize,
```

```
lpStartAddress,
```

```
lpParameter,
```

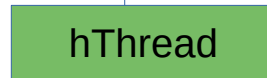
```
dwCreationFlags,
```

```
lpThreadId
```

```
);
```



Inherit ?



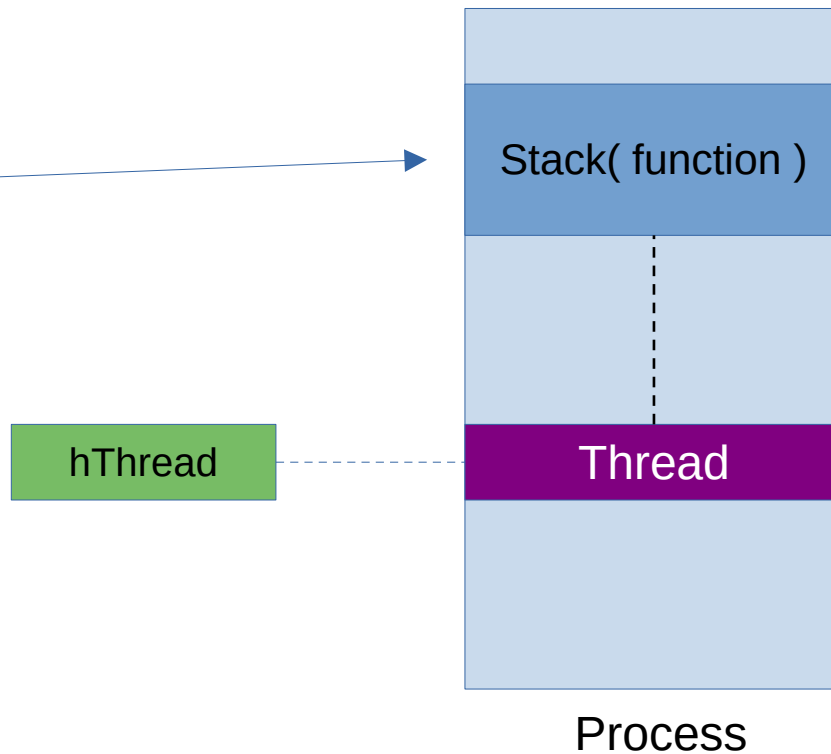
Process

CreateThread prototype

```
HANDLE CreateThread(
```

```
    lpThreadAttributes,  
    dwStackSize,  
    lpStartAddress,  
    lpParameter,  
    dwCreationFlags,  
    lpThreadId
```

```
);
```



CreateThread prototype

HANDLE CreateThread(

lpThreadAttributes,
dwStackSize,
lpStartAddress,
lpParameter,
dwCreationFlags,
lpThreadId

Address of function which we want
to execute by this thread

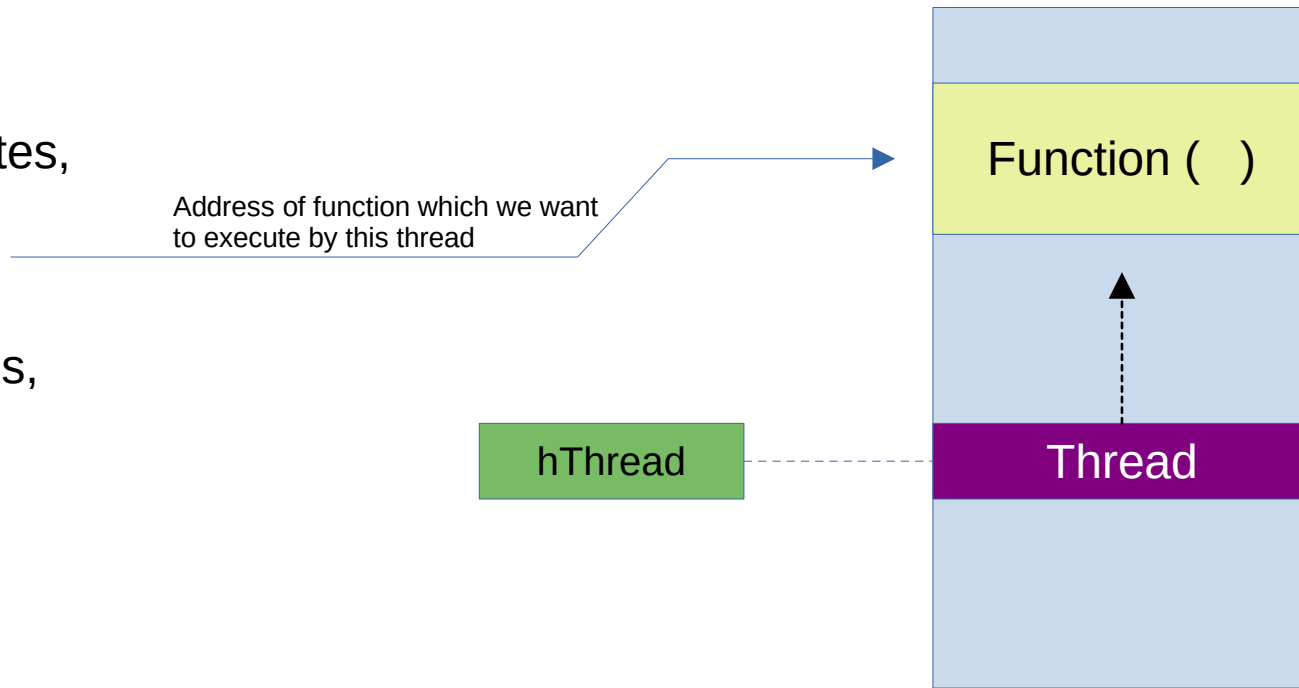
hThread

Function ()

Thread

Process

);

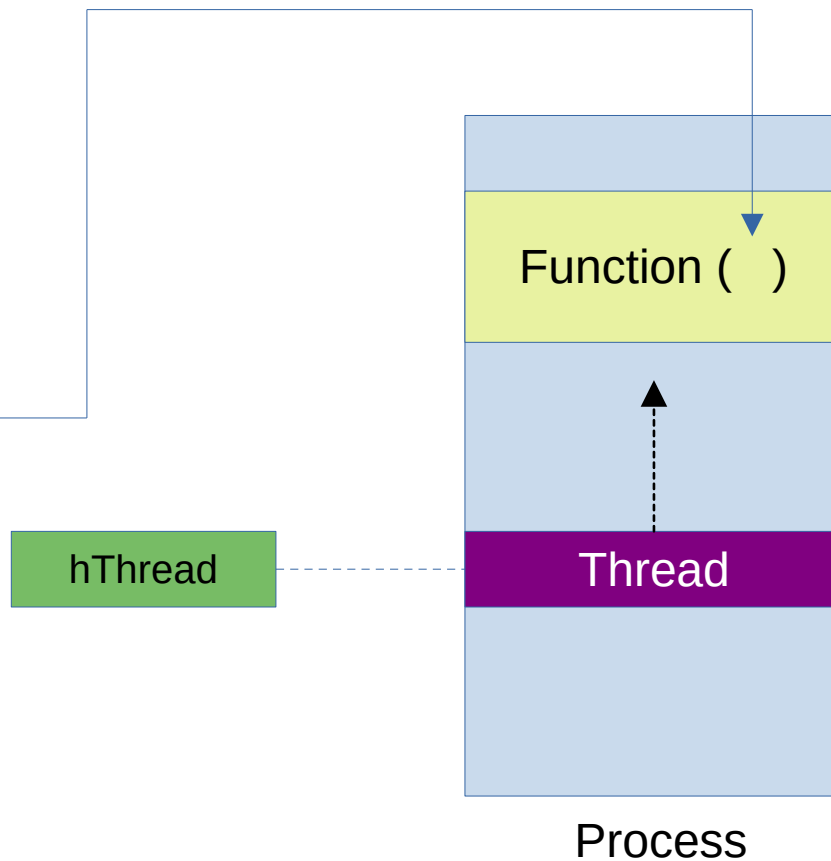


CreateThread prototype

```
HANDLE CreateThread(
```

```
    lpThreadAttributes,  
    dwStackSize,  
    lpStartAddress,  
    lpParameter,  
    dwCreationFlags,  
    lpThreadId
```

```
);
```



CreateThread prototype

```
HANDLE CreateThread(
```

```
    lpThreadAttributes,  
    dwStackSize,  
    lpStartAddress,  
    lpParameter,  
    dwCreationFlags,  
    lpThreadId
```

If we want to create the thread in
suspended state use
CREATE_SUSPENDED Flag else NULL

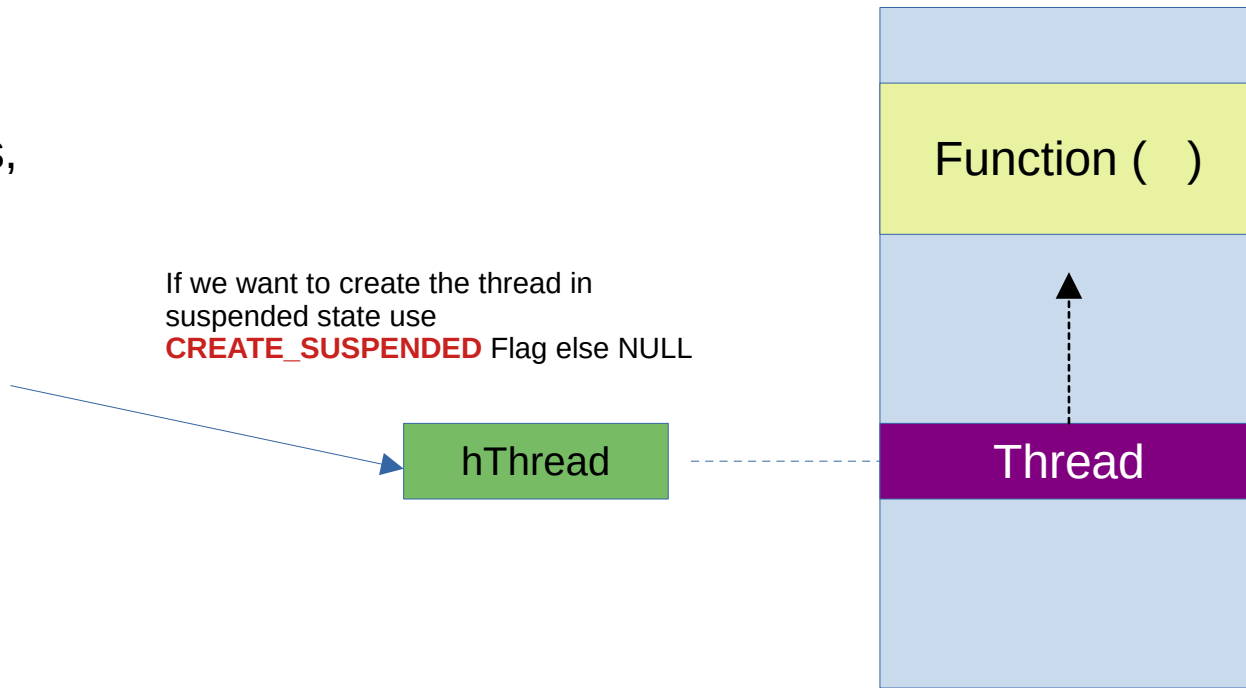
```
);
```

hThread

Function ()

Thread

Process



CreateThread prototype

HANDLE CreateThread(

lpThreadAttributes,
dwStackSize,
lpStartAddress,
lpParameter,
dwCreationFlags,
lpThreadId

);

hThread

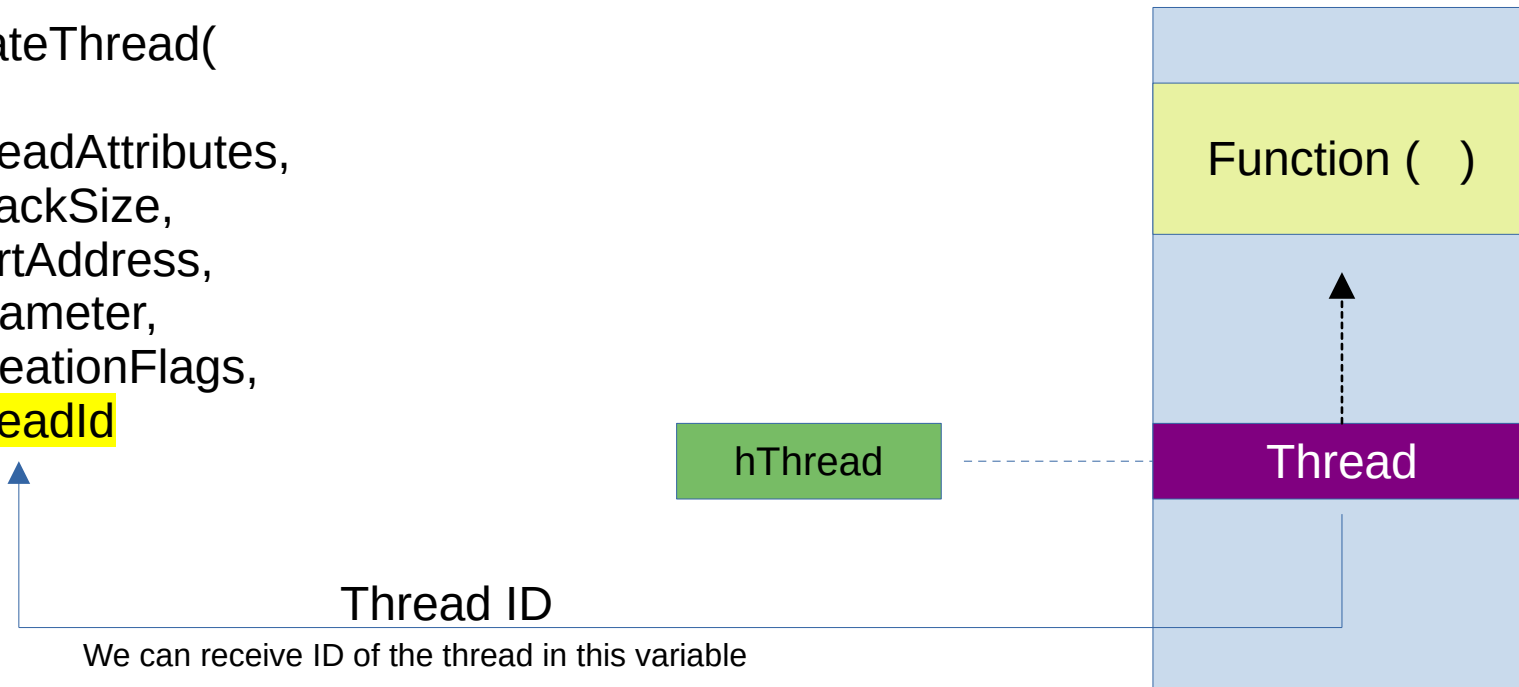
Thread ID

We can receive ID of the thread in this variable

Function ()

Thread

Process



CreateThread() syntax

